c89fa062-afc0-4282-9cdb-b3468c2ec83e

<current>

<request>

You have been given comments on the previous commits you made in the repository.

You are working on an issue in the 'xoseperez/espurna' repository.

<repository_context>
I've cloned the repository in the directory /home/runner/work/espurna/espurna (not in /tmp/inputs). Always use absolute paths when referring to files in the repository.
</repository_context>

<current_datetime>2025-11-26T08:36:25.762Z</current_datetime>

Consider the following problem statement:

<problem_statement>

This section includes details on the stacked pull request you are creating to address the problem above.

<pr_title>
[WIP] Address feedback on relay boot & sync refactoring PR
</pr_title>

<pr_description>
Thanks for the feedback on #2654. I've created this new PR, which merges into #2654, to address your comment. I will work on the changes and keep this PR's description up to date as I make progress.

Original PR: #2654
Triggering review: https://github.com/xoseperez/espurna/pull/

2654#discussion_r2563968377
> @copilot open a new pull request to apply changes based on
<a href="https://github.com/xoseperez/espurna/pull/
2654#discussion_r2563911591">this feedback
</pr_description>

This stacked PR was created to address a specific review comment
on a different pull request.
Focus on addressing the feedback in the review thread below.

Comments on the PR (you are @copilot in this section)

<comments>

<comment_thread>
<file>code/espurna/relay.cpp:2384-2400</file>
<original_commit>b7d5ccf1bcdd7a25d2d13aaf321e2fc404d87c67</
original_commit>

<comment_old>
<author>copilot-pull-request-reviewer[bot]</author>
Logic issue: On line 2388, if a timer exists and contains the
relay ID with the opposite status, the timer is canceled.
However, the variable `timer` is set to `nullptr` after
cancellation, but then line 2399 tries to check and use `timer`
again. Since `timer` is a pointer returned by `find()`,
canceling the timer (which removes it from the list) makes the
pointer dangling. The subsequent check on line 2399 could access
invalid memory. Consider restructuring to avoid using the timer
pointer after cancellation.
</comment_old>

<comment_new>
<comment_id>2563968377</comment_id>
<author>@mcspr</author>
@copilot open a new pull request to apply changes based on
<a href="https://github.com/xoseperez/espurna/pull/
2654#discussion_r2563911591">this feedback
</comment_new>

</comment_thread>

</comments>

The last **1** git commits in this branch are the changes you

have made so far. Use those as your change commit history.
</problem_statement>

Analyze and determine if action or explanation is needed
<rules_for_making_changes>
* Start by fully understanding <problem_statement> and the <comments> before making any changes. You will only address the <comment_new> comments. Reply to new comments as you address them. Ignore comments that are not actionable or directed at you.
* For each new comment, determine:
    - If it a request for changes, a question on the code or a suggestion.
    - Is it addressed to you or someone else, e.g. "Hey @username, ..." or "@username can you ...".
    - Is it a general comment or praise.
* If there are no comments you need to address or reply to, you can stop now without exploring or explaining yourself.
* If the user's ask is not clear, you can reply to the comment and ask for clarification.
* If an action or explanation is needed, follow the steps below.
</rules_for_making_changes>

Ensure that you lint, build and test early and iteratively to validate each code change thoroughly.